

Multiple Choice (10 marks)

1. Why is the organism that causes smallpox not used in the vaccine against smallpox?
 - (a) The organism is extinct in the wild and cannot be harvested for vaccine production
 - (b) The organism that causes smallpox is too costly to use in vaccine production
 - (c) The use of the actual virulent variola virus may be dangerous if too much is administered
 - (d) Smallpox virus is confined to research laboratories
 - (e) There is no specific treatment for smallpox
2. Which of the following is not a form of interspecies interaction?
 - (a) Commensalism
 - (b) Succession
 - (c) Mutualism
 - (d) Parasitism
3. How does the body control the amount of water excreted?
 - (a) The body controls water excretion through sweat glands.
 - (b) Water excretion is determined by the concentration of electrolytes in the body's cells.
 - (c) The kidneys control water excretion by changing the size of the glomerulus.
 - (d) Water excreted is controlled by the hormone insulin.
 - (e) The body adjusts water excretion by altering the pH level of the blood.
4. What problems are posed in referring to the many forms of protozoans as belonging to one single animal phylum?
 - (a) Consistent response to antibiotics across all protozoan forms
 - (b) Inability to adapt to environmental conditions
 - (c) Absence of common ancestral form
 - (d) Extreme diversity, taxonomic organization, possession of certain plant-like features
 - (e) Indistinct cellular structures among different protozoan groups
5. How do cyanobacteria differ from green unicellular algae?
 - (a) Cyanobacteria reduce sulfur compounds.

- (b) Cyanobacteria have no nuclei.
- (c) Green algae cannot reduce nitrogen compounds.
- (d) Green algae cannot photosynthesize.
- (e) Green algae produce cell membranes.

True / False (10 marks)

Answer True or False

- 6. True or False: The correct sequence of membranes in a chloroplast, from innermost to outermost, is thylakoid membrane, inner membrane, outer membrane.
- 7. True or False: Targeting of newly synthesized proteins to the nucleus generally requires two different signal peptides.
- 8. True or False: The frequency of tasters (T) is 0.59 and the frequency of non-tasters (t) is 0.41 among the tested students.
- 9. True or False: Fungi can photosynthesize like plants, converting sunlight into energy and producing glucose in the process.
- 10. True or False: Photosynthesis is not a source of atmospheric carbon, as it utilizes carbon dioxide from the atmosphere.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the mechanisms through which a newborn can contract syphilis, analyzing the potential routes of transmission from mother to child during pregnancy or childbirth.
- 12. Discuss the physiological consequences of severing the dorsal root versus the ventral root of a nerve in a rat's leg, focusing on sensory perception and motor response.

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